

Pankaj Bhagwat

Assistant Professor

Department of Statistics, University of Manitoba, Canada

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Research Interests

Trustworthy AI and Statistical Machine Learning: uncertainty quantification, conformal prediction, fairness, privacy, synthetic data, causal inference, LLMs, and AI safety.

Dependence Modeling and Complex Data: nonparametric dependence measures for metric-space and structured data, and empirical-process methods.

Bayesian and Decision-Theoretic Inference: Bayesian inference, shrinkage methods, and predictive density estimation.

Academic Appointments

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| Sept. 2026–Present |  | Assistant Professor (Tenure-track) , Department of Statistics, University of Manitoba, Canada |
| July–Aug. 2026 |  | Max Wyman Assistant Professor (Limited Term) , Department of Mathematical and Statistical Sciences, University of Alberta, Canada |
| May–Aug. 2026 |  | IVADO Long-Term Visiting Researcher , McGill University, Canada |
| 2023–2026 |  | Postdoctoral Research Fellow , Department of Mathematical and Statistical Sciences, University of Alberta, Canada
Advisors: Bei Jiang and Linglong Kong |

Education

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| 2019–2023 |  | Ph.D. in Statistics, Université de Sherbrooke, Canada
Thesis: <i>On prediction and estimation problems for some multivariate distributions.</i>
Advisor: Éric Marchand. |
| 2014–2019 |  | BS–MS in Mathematics, Indian Institute of Science Education and Research Pune, India
Thesis: <i>Models and Statistical Inference for Multivariate Count Data.</i> |

Publications

Peer-Reviewed Journal Articles & Conference Proceedings

- Bhagwat, P.**, Yang, Z., Wang, Y., Jiang, B., Kong, L. *RSA-CP: Efficient Conformal Prediction in Small-Sample Regimes via Random Score Alignment*. Proceedings of the 43rd International Conference on Machine Learning, 2026.
- Shi, E., **Bhagwat, P.**, Yang, Z., Kong, L., Jiang, B. *Understanding Fairness and Prediction Error through Subspace Decomposition and Influence Analysis*. Proceedings of the 39th Conference on Neural Information Processing Systems, 2025.
- Bhagwat, P.**, Kong, L., Jiang, B. *CBMA: Improving Conformal Prediction through Bayesian Model Averaging*. Proceedings of the 13th International Conference on Learning Representations, 2025.
- Bhagwat, P.**, Marchand, È. *Predictive density estimators with integrated L_1 loss*. Journal of Multivariate Analysis, 197 (2023): 105190.
- Bhagwat, P.**, Marchand, È. *Bayesian inference and prediction for mean-mixtures of normal distributions*. Electronic Journal of Statistics, 17(2) (2023): 1893–1922.
- Bhagwat, P.**, Marchand, È. *On a proper Bayes, but inadmissible estimator*. The American Statistician, 74(3) (2019): 294–296.

Under Review

- 7 **Bhagwat, P.**, Shi, E., Yang, Z., Kong, L., Jiang, B. *Achieving Fairness–Utility Trade-offs through Decoupling Direct and Indirect Bias*. Under review.
- 8 **Bhagwat, P.**, Marchand, È. *Unbiased estimation in one-parameter exponential families for the inverse of the natural parameter with extensions*. Under review.

Preprints

- 9 **Bhagwat, P.**, Strawderman, W., George, E. *Proper Bayes minimax multiple shrinkage estimation*. Preprint.

Selected Talks & Presentations

- 2026  **Can External Information Improve Conformal Prediction without Sacrificing Reliability?** — Université de Montréal, invited talk
- 2026  **Measuring Dependence in Complex Data: A Universal Approach** — Université de Sherbrooke, invited seminar
- 2026  **RSA-CP: Efficient Conformal Prediction in Small-Sample Regimes via Random Score Alignment** — Concordia University, invited talk
- 2026  **Universal Measures of Dependence for Complex Objects** — IVADO Workshop: Statistics in Trustworthy AI, invited talk
- 2026  **Beyond Simple Shrinkage: Optimal Estimation in Multi-Target Settings** — Alberta Mathematics Dialogue, invited talk
- 2025  **CBMA: Improved Conformal Prediction through Bayesian Model Averaging** — International Conference on Learning Representations (ICLR), poster
- 2024  **Integrating Bayesian Model Averaging with Conformal Prediction** — Université de Sherbrooke, invited seminar
- 2024  **Improving Conformal Prediction through Bayesian Model Averaging under Model Misspecification** — CANSSI Showcase, lightning talk
- 2023  **Predictive Density Estimators with Integrated L_1 Loss** — Joint Statistical Meetings (JSM)
- 2023  **On the Construction of Bayes Minimax Multiple Shrinkage Estimators** — Statistical Society of Canada Annual Meeting

Teaching Experience

University of Manitoba

- Fall 2026  STAT 1000 – Basic Statistical Analysis I

University of Alberta

- Fall 2025  STAT 151 – Introduction to Applied Statistics I
- Fall 2024  STAT 151 – Introduction to Applied Statistics I
- Spring 2024  STAT 161 – Introduction to Statistics: Business & Economics

Teaching Assistant, Université de Sherbrooke

- 2020–2023  Applied Statistics; Linear Algebra and Vector Geometry; Vector Calculus; Survival Analysis

Fellowships

- 2022  **ISM Scholarship for Outstanding Ph.D. Candidates**, ISM/UQAM
- 2020–2021  **ISM Fellowship for Graduate Students**, ISM/UQAM
- 2019  **Mitacs Globalink Graduate Fellowship**
- 2018  **Mitacs Globalink Research Internship Fellowship**

- 2017  **Indian Academy of Sciences Summer Research Fellowship**
- 2014–2019  **Department of Science and Technology INSPIRE Fellowship**, Government of India

Awards & Distinctions

- 2026  **Silver Reviewer Award**, International Conference on Machine Learning (ICML)
- 2025  **PDFA Travel Award for ICLR**, University of Alberta
- 2024  **Dean's Honor List**, Faculty of Sciences, Université de Sherbrooke
- 2021, 2023  **Student Travel Award**, Statistical Society of Canada
- 2019  **Gold Medal**, BS–MS Program, IISER Pune

Professional Service

-  **Reviewer.** Journal of the American Statistical Association; Statistica Sinica; Journal of Multivariate Analysis; Data Science in Science; Transactions on Machine Learning Research; ICLR; ICML; NeurIPS; AAAI.
- 2025  **Judge.** University of Alberta Three Minute Thesis (3MT®) Competition.